

```
In [1]: #import libraries  
!pip install pandas numpy matplotlib seaborn scikit-learn
```

Requirement already satisfied: pandas in /Users/roshan/anaconda3/lib/python3.11/site-packages (2.0.3)
 Requirement already satisfied: numpy in /Users/roshan/anaconda3/lib/python3.11/site-packages (2.4.2)
 Requirement already satisfied: matplotlib in /Users/roshan/anaconda3/lib/python3.11/site-packages (3.7.2)
 Requirement already satisfied: seaborn in /Users/roshan/anaconda3/lib/python3.11/site-packages (0.12.2)
 Requirement already satisfied: scikit-learn in /Users/roshan/anaconda3/lib/python3.11/site-packages (1.3.0)
 Requirement already satisfied: python-dateutil<=2.8.2 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from pandas) (2.8.2)
 Requirement already satisfied: pytz<=2020.1 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from pandas) (2025.2)
 Requirement already satisfied: tzdata<=2022.1 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from pandas) (2023.3)
 Requirement already satisfied: contourpy<=1.0.1 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from matplotlib) (1.0.5)
 Requirement already satisfied: cycler<=0.10 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from matplotlib) (0.11.0)
 Requirement already satisfied: fonttools<=4.22.0 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from matplotlib) (4.25.0)
 Requirement already satisfied: kiwisolver<=1.0.1 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from matplotlib) (1.4.4)
 Requirement already satisfied: packaging<=20.0 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from matplotlib) (23.1)
 Requirement already satisfied: pillow<=6.2.0 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from matplotlib) (9.4.0)
 Requirement already satisfied: pyparsing<3.1,>=2.3.1 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from matplotlib) (3.0.9)
 Requirement already satisfied: scipy<=1.5.0 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from scikit-learn) (1.11.1)
 Requirement already satisfied: joblib<=1.1.1 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from scikit-learn) (1.2.0)
 Requirement already satisfied: threadpoolctl<=2.0.0 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from scikit-learn) (2.2.0)
 Requirement already satisfied: six<=1.5 in /Users/roshan/anaconda3/lib/python3.11/site-packages (from python-dateutil<=2.8.2->pandas) (1.16.0)

Collecting numpy

Obtaining dependency information for numpy from https://files.pythonhosted.org/packages/1a/2e/151484f49fd03944c4a3ad9c418ed193cfd02724e138ac8a9505d056c582/numpy-1.26.4-cp311-cp311-macosx_11_0_arm64.whl.metadata

Downloading numpy-1.26.4-cp311-cp311-macosx_11_0_arm64.whl.metadata (114 kB)

114.8/114.8 kB 2.3 MB/s eta 0:00:00 MB/s eta 0:00:01

Downloading numpy-1.26.4-cp311-cp311-macosx_11_0_arm64.whl (14.0 MB)

14.0/14.0 MB 28.5 MB/s eta 0:00:00m eta 0:00:01[36m0:00:01

Installing collected packages: numpy

Attempting uninstall: numpy

Found existing installation: numpy 2.4.2

Uninstalling numpy-2.4.2:

Successfully uninstalled numpy-2.4.2

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

tables 3.8.0 requires blosc2<=2.0.0, which is not installed.

tables 3.8.0 requires cython<=0.29.21, which is not installed.

gensim 4.3.0 requires FuzzyTM<=0.4.0, which is not installed.

numba 0.57.1 requires numpy<1.25,>=1.21, but you have numpy 1.26.4 which is incompatible.

opencv-python 4.13.0.92 requires numpy<=2; python_version >= "3.9", but you

have numpy 1.26.4 which is incompatible.

Successfully installed numpy-1.26.4

```
In [5]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn
import sqlite3

print("All libraries imported successfully!")
```

All libraries imported successfully!

```
In [ ]: # import database
```

```
In [13]: import sqlite3
import pandas as pd

# Connect to database
conn = sqlite3.connect("customer_churn.db")

# Get all table names
sql_query = """
SELECT name
FROM sqlite_master
WHERE type = 'table'
"""

tables = pd.read_sql(sql_query, conn)

# Create dataframe for each table
for table_name in tables["name"]:
    df = pd.read_sql(f"SELECT * FROM {table_name}", conn)
    globals()[f"df_{table_name}"] = df
    print(f"Created dataframe: df_{table_name}")

# Close database connection
conn.close()
```

Created dataframe: df_db_customer
Created dataframe: df_db_subscription
Created dataframe: df_db_support

```
In [14]: import sqlite3
import pandas as pd

# Connect to database
conn = sqlite3.connect("customer_churn.db")

# Get all table names
sql_query = """
SELECT name
FROM sqlite_master
WHERE type = 'table'
"""

tables = pd.read_sql(sql_query, conn)

# Print table names and column names
for table_name in tables["name"]:

    df = pd.read_sql(f"SELECT * FROM {table_name}", conn)
```

```

globals()[f"df_{table_name}"] = df

print("\n=====")
print("Table Name:", table_name)
print("Columns:")

for column in df.columns:
    print(" -", column)

# Close connection
conn.close()

```

```
=====
```

```
Table Name: db_customer
```

```
Columns:
```

- customerid
- name
- country
- state
- gender
- dob
- interests
- pincode

```
=====
```

```
Table Name: db_subscription
```

```
Columns:
```

- customerid
- subscription_start_date
- subscription_type
- renewal_date
- plan_type
- contract_type
- cancellation_date
- cancellation_reason
- monthly_charges
- cltv
- churn_score

```
=====
```

```
Table Name: db_support
```

```
Columns:
```

- customerid
- complaint_date
- escalations
- csat_score
- col_1
- comment

```
In [ ]: # data cleaning
```

```
In [20]: df_db_customer.head()
#df_db_customer.tail()
```

Out [20]:

	customerid	name	country	state	gender	dob	interests	pincode
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00	travel	None
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00	None	None
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00	movie	None
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00	None	None
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00	drama	None

In [16]: df_db_customer.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      21 non-null    object
1   name            21 non-null    object
2   country         18 non-null    object
3   state           21 non-null    object
4   gender          21 non-null    object
5   dob             21 non-null    object
6   interests       4 non-null     object
7   pincode         0 non-null     object
dtypes: object(8)
memory usage: 1.4+ KB
```

```
In [ ]: # i have to drop two columns- interest and pincode
# rename column-name
#change datatype -dob
# data standarddization-gender
#fix missing value
```

In [28]: # rename column-name

```
df_db_customer.rename(columns={'name': 'customer_name'}, inplace=True)
df_db_customer
```

Out [28] :

	customerid	customer_name	country	state	gender	dob	interests	pincode
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00	travel	None
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00	None	None
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00	movie	None
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00	None	None
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00	drama	None
5	0013-MHZWF	durga	None	Delhi	Women	1988-12-10 00:00:00	None	None
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21 00:00:00	None	None
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14 00:00:00	None	None
8	0015-UOCOJ	maya	None	Kathmandu	Women	1985-07-07 00:00:00	None	None
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29 00:00:00	None	None
10	0017-DINOC	shiva	India	Maharashtra	Men	1997-01-22 00:00:00	None	None
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18 00:00:00	None	None
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01 00:00:00	None	None
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25 00:00:00	None	None
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Men	1979-11-11 00:00:00	None	None
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28 00:00:00	job	None
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00	None	None

	customerid	customer_name	country	state	gender	dob	interests	pincode
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00	None	None
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00	None	None
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00	None	None
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00	None	None

```
In [ ]: #drop column
df_db_customer.drop(columns=['interests', 'pincode'], inplace=True)
```

```
In [55]: print(df_db_customer.columns.tolist())

['customerid', 'customer_name', 'country', 'state', 'gender', 'dob']
```

```
In [56]: df_db_customer
```

```
Out[56]:
```

	customerid	customer_name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00
5	0013-MHZWF	durga	None	Delhi	Women	1988-12-10 00:00:00
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21 00:00:00
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14 00:00:00
8	0015-UOCOJ	maya	None	Kathmandu	Women	1985-07-07 00:00:00
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29 00:00:00
10	0017-DINOC	shiva	India	Maharashtra	Men	1997-01-22 00:00:00
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18 00:00:00
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01 00:00:00
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25 00:00:00
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Men	1979-11-11 00:00:00
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28 00:00:00
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00

In [58]: `df_db_customer.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      21 non-null    object
1   customer_name   21 non-null    object
2   country         18 non-null    object
3   state           21 non-null    object
4   gender          21 non-null    object
5   dob             21 non-null    object
dtypes: object(6)
memory usage: 1.1+ KB
```

In [63]: `#change datatype -dob`
`df_db_customer['dob'] = pd.to_datetime(df_db_customer['dob'])`

In [64]: `df_db_customer`

Out[64]:

	customerid	customer_name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05
5	0013-MHZWF	durga	None	Delhi	Women	1988-12-10
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14
8	0015-UOCOJ	maya	None	Kathmandu	Women	1985-07-07
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29
10	0017-DINOC	shiva	India	Maharashtra	Men	1997-01-22
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Men	1979-11-11
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06

In [65]: `df_db_customer.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      21 non-null    object
1   customer_name   21 non-null    object
2   country         18 non-null    object
3   state           21 non-null    object
4   gender           21 non-null    object
5   dob             21 non-null    datetime64[ns]
dtypes: datetime64[ns](1), object(5)
memory usage: 1.1+ KB
```

```
In [73]: # data standarddization-gender
#df_db_customer['gender'].unique()

df_db_customer['gender'].replace({'Men':'Male', 'Women':'Female:'})
```

```
Out[73]: 0      Male
1      Male
2     Female
3      Male
4     Female
5     Female:
6     Female
7      Male
8     Female:
9      Male
10     Male
11     Female
12     Female
13     Female
14      Male
15     Female
16     Female
17     Female
18      Male
19      Male
20     Female:
Name: gender, dtype: object
```

```
In [74]: df_db_customer['gender'].unique()
```

```
Out[74]: array(['Male', 'Female', 'Female:'], dtype=object)
```

```
In [75]: df_db_customer
```

Out [75]:

	customerid	customer_name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05
5	0013-MHZWF	durga	None	Delhi	Female:	1988-12-10
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14
8	0015-UOCOJ	maya	None	Kathmandu	Female:	1985-07-07
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29
10	0017-DINOC	shiva	India	Maharashtra	Male	1997-01-22
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Male	1979-11-11
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Male	1991-05-14
20	0023-UYUPN	sudevi	India	Maharashtra	Female:	1977-10-06

In [76]: *#fix missing value-country*

df_db_customer['country'].isna()

Out [76]:

```

0    False
1    False
2    False
3    False
4    False
5     True
6    False
7    False
8     True
9    False
10   False
11   False
12    True
13   False
14   False
15   False
16   False
17   False
18   False
19   False
20   False
Name: country, dtype: bool

```

```
In [77]: df_db_customer[df_db_customer['country'].isna()]
```

```
Out[77]:
```

	customerid	customer_name	country	state	gender	dob
5	0013-MHZWF	durga	None	Delhi	Female:	1988-12-10
8	0015-UOCOJ	maya	None	Kathmandu	Female:	1985-07-07
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01

```
In [82]: #df_db_customer[['country' , 'state']]
#country and state - unique value pair

state_country_mapping = df_db_customer.dropna(subset=['country']).set_index('country')
df_db_customer['country']=df_db_customer['country'].fillna(df_db_customer['state_country_mapping'])
```

```
In [84]: df_db_customer[df_db_customer['country'].isna()]
```

```
Out[84]:
```

	customerid	customer_name	country	state	gender	dob
--	------------	---------------	---------	-------	--------	-----

```
In [86]: df_db_customer[['country', 'state']]
```

```
Out[86]:
```

	country	state
0	India	Maharashtra
1	India	Karnataka
2	India	Delhi
3	India	Nagaland
4	India	Delhi
5	India	Delhi
6	India	Meghalaya
7	India	Rajasthan
8	Nepal	Kathmandu
9	Nepal	Kathmandu
10	India	Maharashtra
11	India	Karnataka
12	India	Telangana
13	India	Meghalaya
14	India	Uttar Pradesh
15	India	Delhi
16	India	Meghalaya
17	India	Rajasthan
18	India	Telangana
19	India	Uttar Pradesh
20	India	Maharashtra

```
In [87]: df_db_customer
```

Out [87]:

	customerid	customer_name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05
5	0013-MHZWF	durga	India	Delhi	Female:	1988-12-10
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14
8	0015-UOCOJ	maya	Nepal	Kathmandu	Female:	1985-07-07
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29
10	0017-DINOC	shiva	India	Maharashtra	Male	1997-01-22
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18
12	0018-NYROU	chitra	India	Telangana	Female	2004-12-01
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Male	1979-11-11
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Male	1991-05-14
20	0023-UYUPN	sudevi	India	Maharashtra	Female:	1977-10-06

In [88]: df_db_subscription.head()

Out [88]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	1
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	1

In [89]: df_db_subscription.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customerid                           21 non-null     object
1   subscription_start_date              21 non-null     object
2   subscription_type                     21 non-null     object
3   renewal_date                         21 non-null     object
4   plan_type                            21 non-null     object
5   contract_type                        21 non-null     object
6   cancellation_date                    6 non-null      object
7   cancellation_reason                  6 non-null      object
8   monthly_charges                      21 non-null     float64
9   cltv                                 21 non-null     int64
10  churn_score                           21 non-null     int64
dtypes: float64(1), int64(2), object(8)
memory usage: 1.9+ KB
```

```
In [91]: date_col=['subscription_start_date','renewal_date','cancellation_date']
df_db_subscription[date_col]=df_db_subscription[date_col].apply(pd.to_datetime)
df_db_subscription.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customerid                           21 non-null     object
1   subscription_start_date              21 non-null     datetime64[ns]
2   subscription_type                     21 non-null     object
3   renewal_date                         21 non-null     datetime64[ns]
4   plan_type                            21 non-null     object
5   contract_type                        21 non-null     object
6   cancellation_date                    6 non-null      datetime64[ns]
7   cancellation_reason                  6 non-null      object
8   monthly_charges                      21 non-null     float64
9   cltv                                 21 non-null     int64
10  churn_score                           21 non-null     int64
dtypes: datetime64[ns](3), float64(1), int64(2), object(5)
memory usage: 1.9+ KB
```

```
In [92]: df_db_support.head()
```

```
Out[92]:
```

	customerid	complaint_date	escalations	csat_score	col_1	comment
0	0003-MKNFE	2024-08-28 00:00:00	N	60	None	service issue
1	0003-MKNFE	2024-08-28 00:00:00	Y	10	None	demaned refund
2	0013-EXCHZ	2024-01-20 00:00:00	Y	20	None	None
3	0013-MHZWF	2025-03-18 00:00:00	N	90	None	guidance to renew
4	0013-SMEOE	2024-11-01 00:00:00	N	30	None	None

```
In [93]: df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      9 non-null     object
1   complaint_date  9 non-null     object
2   escalations     9 non-null     object
3   csat_score      9 non-null     int64
4   col_1           0 non-null     object
5   comment         4 non-null     object
dtypes: int64(1), object(5)
memory usage: 564.0+ bytes
```

```
In [94]: df_db_support.drop(columns=['col_1', 'comment'], inplace=True)
```

```
In [95]: #df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      9 non-null     object
1   complaint_date  9 non-null     object
2   escalations     9 non-null     object
3   csat_score      9 non-null     int64
dtypes: int64(1), object(3)
memory usage: 420.0+ bytes
```

```
In [97]: df_db_support['complaint_date '] = pd.to_datetime(df_db_support['complaint_date '])
df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      9 non-null     object
1   complaint_date  9 non-null     object
2   escalations     9 non-null     object
3   csat_score      9 non-null     int64
4   complaint_date  9 non-null     datetime64[ns]
dtypes: datetime64[ns](1), int64(1), object(3)
memory usage: 492.0+ bytes
```

```
In [ ]: # feature engineering and data analysis
```

```
In [102]: #create a new column using existing col- churn flag
df_db_subscription['churn_flag'] = np.where(df_db_subscription['cancellation'] == 'Completed', 1, 0)
df_db_subscription.head()
```

Out[102]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contra
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	

In [109... *#first fix support table duplicates then merge*

```
df=(df_db_subscription
      .merge(df_db_customer,on = 'customerid',how='left')
      .merge(df_db_support,on = 'customerid',how='left') )
```

In [112... df_db_subscription.shape

Out[112]: (21, 12)

In [111... df.shape

Out[111]: (23, 21)

In [113... df.head()

Out[113]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contra
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
3	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	
4	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	

5 rows × 21 columns

In [114... df_db_subscription['customerid'].nunique()

Out[114]: 21

In [115... df_db_customer['customerid'].nunique()

Out[115]: 21

In [116... df_db_support['customerid'].nunique()

Out[116]: 7

```
In [117... df_db_support['customerid'].size
```

```
Out[117]: 9
```

```
In [119... df_db_support
```

```
Out[119]:
```

	customerid	complaint_date	escalations	csat_score	complaint_date
0	0003-MKNFE	2024-08-28 00:00:00	N	60	2024-08-28
1	0003-MKNFE	2024-08-28 00:00:00	Y	10	2024-08-28
2	0013-EXCHZ	2024-01-20 00:00:00	Y	20	2024-01-20
3	0013-MHZWF	2025-03-18 00:00:00	N	90	2025-03-18
4	0013-SMEOE	2024-11-01 00:00:00	N	30	2024-11-01
5	0017-IUDMW	2024-04-10 00:00:00	Y	25	2024-04-10
6	0019-EFAEP	2024-09-27 00:00:00	Y	30	2024-09-27
7	0022-TCJCI	2024-09-13 00:00:00	Y	10	2024-09-13
8	0022-TCJCI	2024-09-14 00:00:00	N	90	2024-09-14

```
In [124... df_db_support
```

```
Out[124]:
```

	customerid	complaint_date	escalations	csat_score	complaint_date
0	0003-MKNFE	2024-08-28 00:00:00	N	60	2024-08-28
1	0003-MKNFE	2024-08-28 00:00:00	Y	10	2024-08-28
2	0013-EXCHZ	2024-01-20 00:00:00	Y	20	2024-01-20
3	0013-MHZWF	2025-03-18 00:00:00	N	90	2025-03-18
4	0013-SMEOE	2024-11-01 00:00:00	N	30	2024-11-01
5	0017-IUDMW	2024-04-10 00:00:00	Y	25	2024-04-10
6	0019-EFAEP	2024-09-27 00:00:00	Y	30	2024-09-27
7	0022-TCJCI	2024-09-13 00:00:00	Y	10	2024-09-13
8	0022-TCJCI	2024-09-14 00:00:00	N	90	2024-09-14

```
In [126... df_db_support['complaint_count']=df_db_support.groupby('customerid')['custor
```

```
In [128... df_db_support=df_db_support.sort_values('complaint_date').drop_duplicates('c
```

```
In [129... df_db_support['customerid'].size
```

```
Out[129]: 7
```

```
In [130... #first fix support table duplicates then merge
df=(df_db_subscription
    .merge(df_db_customer,on = 'customerid',how='left')
    .merge(df_db_support,on = 'customerid',how='left') )
```

```
In [131... df.shape
```

```
Out[131]: (21, 22)
```

```
In [133... df.to_csv('exported_churn_data.csv', index=False)
```

```
In [134... df.columns
```

```
Out[134]: Index(['customerid', 'subscription_start_date', 'subscription_type',
        'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
        'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
        'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
        'complaint_date', 'escalations', 'csat_score', 'complaint_date ',
        'complaint_count'],
        dtype='object')
```

```
In [140... churn_rate=df['churn_flag'].mean()*100
print("churn Rate = ", round(churn_rate, 2), "%")
```

churn Rate = 28.57 %

```
In [143... #retention rate
retention_rate = 100-churn_rate
print("retention_rate = ", round(retention_rate, 2), "%")
```

retention_rate = 71.43 %

```
In [145... df.head(5)
```

```
Out[145]:
```

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contra
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	

5 rows x 22 columns

```
In [149... #churn by plan type
churn_by_plan=(df.groupby('plan_type')['churn_flag'].mean().mul(100).round(2))
print(churn_by_plan)
```

```
plan_type  churn_rate_pct
0    Basic             60.00
1    Premium            14.29
2    Standard           22.22
```

```
In [153... churn_by_state = (
    df.groupby('state')
      .agg(
        churn_rate_pct=('churn_flag', 'mean'),
        total_revenue=('monthly_charges', 'sum'),
        user_count=('customerid', 'nunique')
      )
      .reset_index()
)
```

```
churn_by_state['churn_rate_pct'] = (
    churn_by_state['churn_rate_pct'] * 100
).round(2)

print(churn_by_state)
```

	state	churn_rate_pct	total_revenue	user_count
0	Delhi	25.00	52.96	4
1	Karnataka	100.00	20.98	2
2	Kathmandu	0.00	20.98	2
3	Maharashtra	0.00	50.97	3
4	Meghalaya	66.67	42.97	3
5	Nagaland	0.00	22.99	1
6	Rajasthan	0.00	36.98	2
7	Telangana	50.00	30.98	2
8	Uttar Pradesh	0.00	115.98	2

```
In [152... print(df.columns.tolist())
```

```
['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score', 'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score', 'complaint_date ', 'complaint_count']
```

```
In [154... #churn_by_subscription
churn_by_subscription = (
    df.groupby('subscription_type')
    .agg(
        churn_rate_pct=('churn_flag', 'mean'),
        total_revenue=('monthly_charges', 'sum'),
        user_count=('customerid', 'nunique')
    )
    .reset_index()
)

churn_by_subscription['churn_rate_pct'] = (
    churn_by_subscription['churn_rate_pct'] * 100
).round(2)

print(churn_by_subscription)
```

	subscription_type	churn_rate_pct	total_revenue	user_count
0	Organic	0.00	145.91	9
1	Paid	16.67	174.94	6
2	Refferal	83.33	74.94	6

```
In [158... # ARPU- AVG REVENUE PER USER

arpu=df['monthly_charges'].mean()
print('ARPU = ',round(arpu,2))
```

```
ARPU = 18.85
```

```
In [165... #average customer tenure
#count of days has used our services :cancellation date else current date
today = pd.Timestamp.today()

df['tenure_days'] = np.where(
    df['cancellation_date'].notna(),
    (df['cancellation_date'] - df['subscription_start_date']).dt.days,
    (today - df['subscription_start_date']).dt.days
)

avg_tenure = df['tenure_days'].mean()
```

```
print("Average tenure (days) =", round(avg_tenure, 0))
```

Average tenure (days) = 1520.0

```
In [170... # revenue at risk -revenue lost from churn users

revenue_at_risk = df.loc[df['churn_flag'] == 1, 'monthly_charges'].sum()

print("Revenue at Risk (Monthly) Rs ('k')=", round(revenue_at_risk, 2))
```

Revenue at Risk (Monthly) Rs ('k')= 73.94

```
In [171... # escalation Rate

escalation_rate = (df['escalations'] == 'Y').mean() * 100

print("Escalation Rate =", round(escalation_rate, 2), "%")
```

Escalation Rate = 19.05 %

```
In [173... # average complainint per user

avg_complaints_per_user = df['complaint_count'].sum()/df['customerid'].nunique()

print("Average Complaints per User =", round(avg_complaints_per_user, 2))
```

Average Complaints per User = 0.43

```
In [175... # correlation escalation vs churn

df['escalation_flag'] = np.where(df['escalations'] == 'Y', 1, 0)

correlation = df['escalation_flag'].corr(df['churn_flag'])

print("Correlation (Escalation vs Churn) =", round(correlation, 2))
```

Correlation (Escalation vs Churn) = 0.77

```
In [177... df['churn_score'].head()
```

```
Out[177]: 0    12
1    91
2    34
3     8
4    88
Name: churn_score, dtype: int64
```

```
In [180... # churn risk - create a column using a existing column

conditions = [
    (df['churn_score'] < 50),
    (df['churn_score'] >= 50) & (df['churn_score'] < 70),
    (df['churn_score'] >= 70)
]

choices = ['low', 'med', 'high']

df['churn_risk'] = np.select(
    conditions,
    choices,
    default='unknown'
)
```

```
In [181... df[['churn_score', 'churn_risk']].head()
```

Out[181]:

	churn_score	churn_risk
0	12	low
1	91	high
2	34	low
3	8	low
4	88	high

	churn_score	churn_risk
0	12	low
1	91	high
2	34	low
3	8	low
4	88	high

In [183... df.columns

Out[183]: Index(['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score', 'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score', 'complaint_date ', 'complaint_count', 'tenure_days', 'escalation_flag', 'churn_risk'], dtype='object')

In [188... df_visual = df.copy()

In [195... df_visual.columns

Out[195]: Index(['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score', 'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score', 'complaint_date ', 'complaint_count', 'tenure_days', 'escalation_flag', 'churn_risk'], dtype='object')

In [215... *#monthly churn trend (time series kpi)*

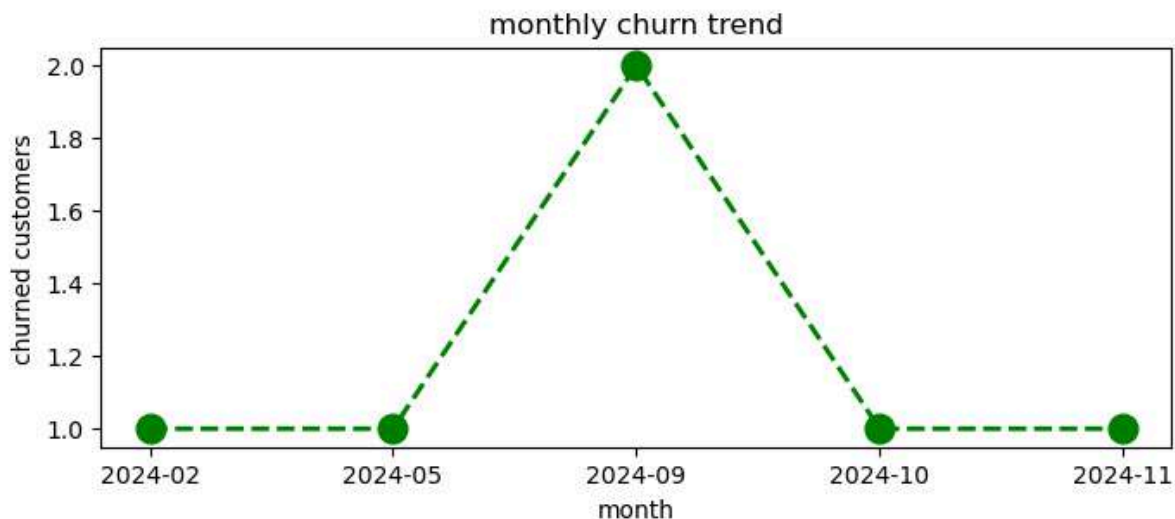
```
df_visual['cancellation_month'] = df_visual['cancellation_date'].dt.to_timestamp()

churn_trend = df_visual[df_visual['churn_flag'] == 1].groupby('cancellation_month').sum()

churn_trend_values = churn_trend.values
plt.figure(figsize=(8,3))

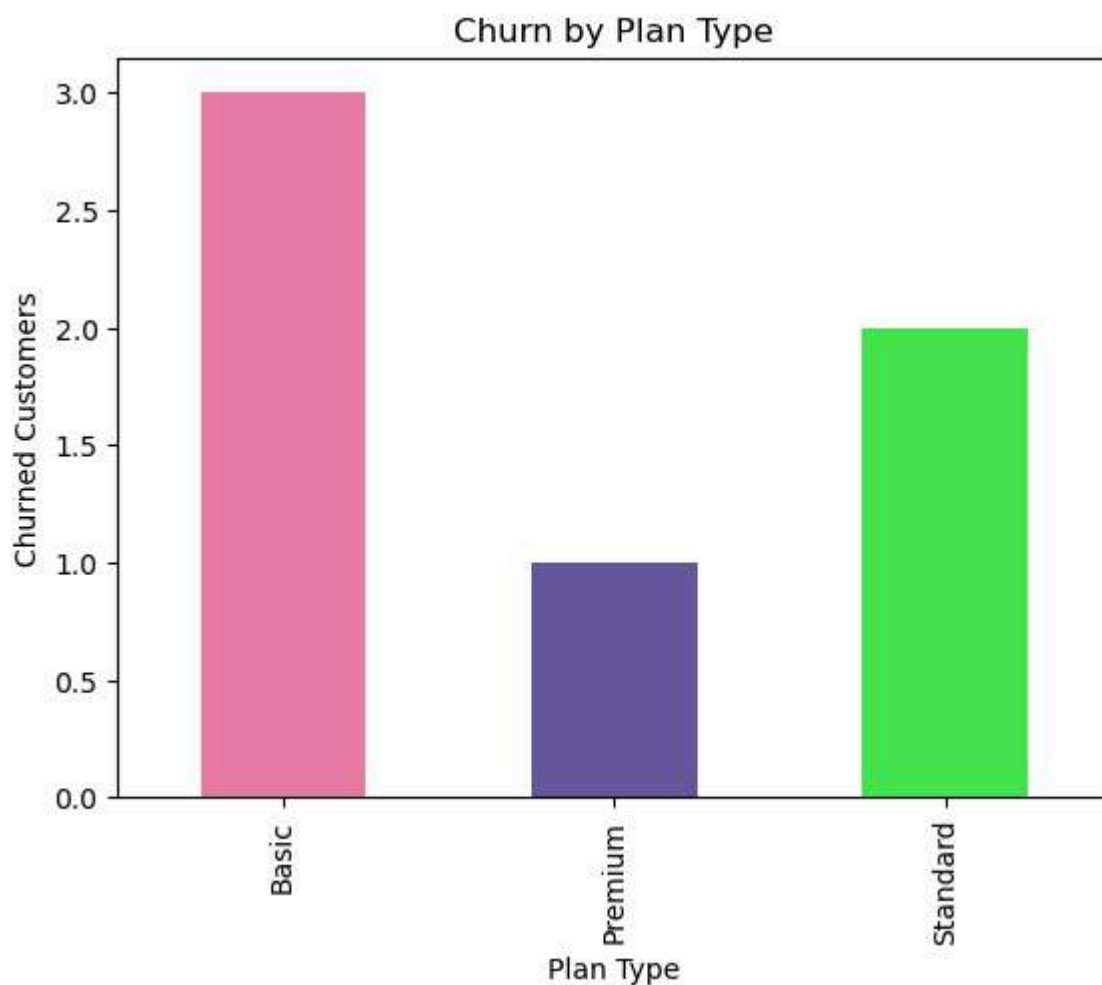
plt.plot(churn_trend.index.astype(str), churn_trend_values,
         color='green', marker='o', linestyle='dashed',
         linewidth=2, markersize=12)

plt.title('monthly churn trend')
plt.xlabel('month')
plt.ylabel('churned customers')
plt.show()
```



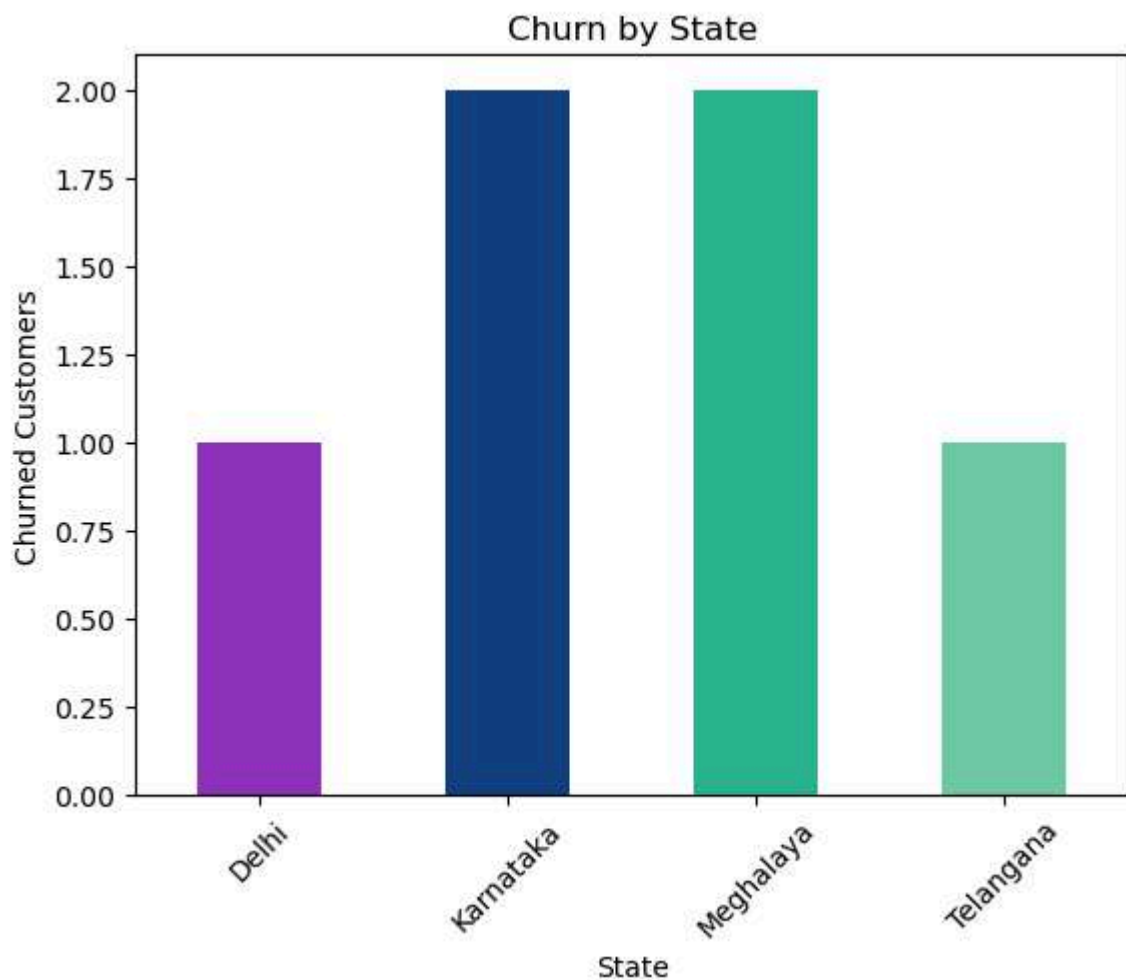
```
In [224]: churn_by_plan = df_visual[df_visual['churn_flag']==1].groupby('plan_type').size()
churn_by_plan.plot(kind='bar', color=np.random.rand(len(churn_by_plan), 3))

plt.title('Churn by Plan Type')
plt.xlabel('Plan Type')
plt.ylabel('Churned Customers')
plt.show()
```



```
In [225]: # churn by states
churn_by_state = df_visual[df_visual['churn_flag']==1].groupby('state').size()
churn_by_state.plot(kind='bar', color=np.random.rand(len(churn_by_state), 3))
```

```
plt.title('Churn by State')
plt.xlabel('State')
plt.ylabel('Churned Customers')
plt.xticks(rotation=45)
plt.show()
```



```
In [ ]: # visualization using seaborn
```

```
In [226... # encoding- convert str to numeric so that we can find the correlation between
df_visual.columns
```

```
Out[226]: Index(['customerid', 'subscription_start_date', 'subscription_type',
               'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
               'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
               'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
               'complaint_date', 'escalations', 'csat_score', 'complaint_date ',
               'complaint_count', 'tenure_days', 'escalation_flag', 'churn_risk',
               'cancellation_month'],
              dtype='object')
```

```
In [230... df_visual[[
               'plan_type', 'contract_type',
               'churn_score',
               'churn_flag',
               'escalations',
               'churn_risk'
            ]].head()
```

Out [230]:

	plan_type	contract_type	churn_score	churn_flag	escalations	churn_risk
0	Standard	Annual	12	0	NaN	low
1	Premium	Annual	91	1	Y	high
2	Basic	Monthly	34	0	NaN	low
3	Premium	Annual	8	0	NaN	low
4	Standard	Monthly	88	1	Y	high

In [233... df_encoded.head()

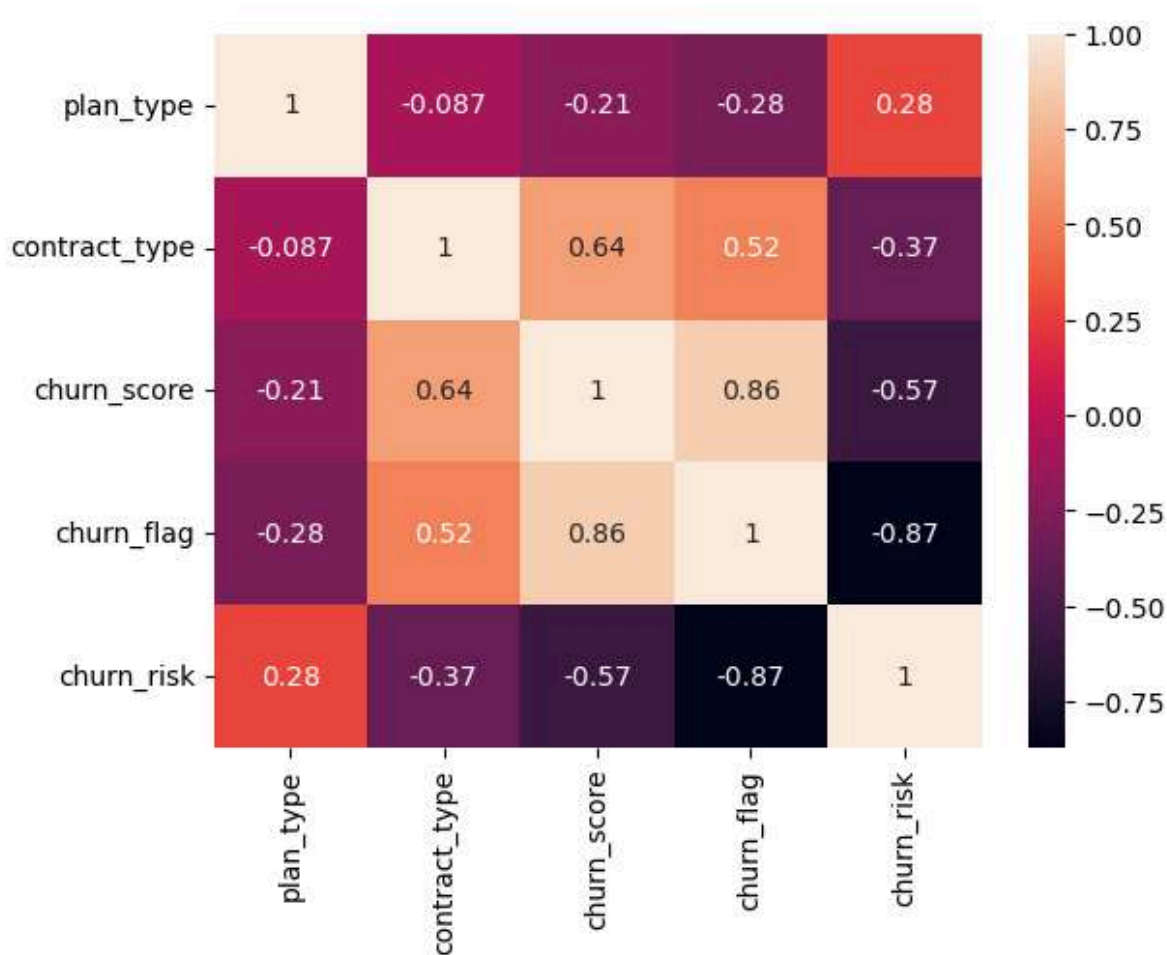
Out [233]:

	plan_type	contract_type	churn_score	churn_flag	escalations	churn_risk
0	2	0	12	0	NaN	1
1	1	0	91	1	Y	0
2	0	1	34	0	NaN	1
3	1	0	8	0	NaN	1
4	2	1	88	1	Y	0

In [235... `import warnings`
`warnings.filterwarnings("ignore")`

In [242... `# incorrect method`
`df_encoded = df_visual[['plan_type', 'contract_type',`
`'churn_score',`
`'churn_flag',`
`'churn_risk',`
`'escalations']]`
`categorical_cols = ['plan_type', 'contract_type', 'churn_risk']`
`for col in categorical_cols:`
`df_encoded[col] = df_encoded[col].astype('category').cat.codes`

In [241... `#heatmap (correlation matrix)`
`sns.heatmap(df_encoded.corr(numeric_only=True), annot=True)`
`plt.show()`



```
In [253...] df_encoded = df_visual[
    ['plan_type', 'contract_type', 'churn_score',
     'churn_flag', 'churn_risk', 'escalations']
].copy()

order_mappings = {
    'plan_type': ['Basic', 'Standard', 'Premium'],
    'contract_type': ['Monthly', 'Annual'],
    'churn_risk': ['low', 'medium', 'high']
}

for col, order in order_mappings.items():
    df_encoded[col] = pd.Categorical(
        df_encoded[col], categories=order, ordered=True
    ).codes

df_encoded['escalations'] = df_encoded['escalations'].replace(
    {'N':0, 'Y':1}
).fillna(0).astype(int)
```

```
In [257...] df_encoded.head()
```

```
Out[257]:
```

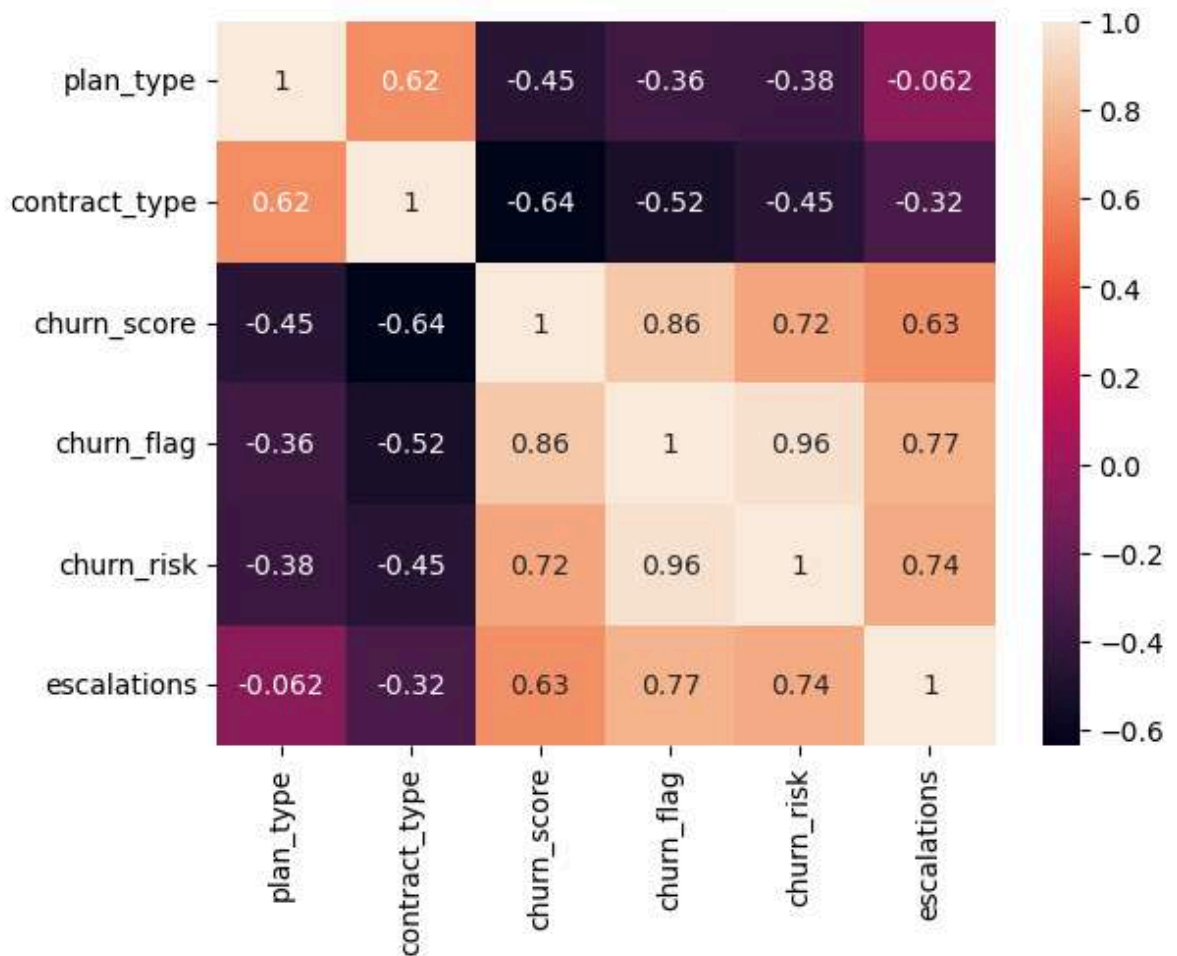
	plan_type	contract_type	churn_score	churn_flag	churn_risk	escalations
0	1	1	12	0	0	0
1	2	1	91	1	2	1
2	0	0	34	0	0	0
3	2	1	8	0	0	0
4	1	0	88	1	2	1

```
In [261... df_encoded[['plan_type', 'contract_type', 'churn_risk', 'escalations']].head()
```

```
Out[261]:
```

	plan_type	contract_type	churn_risk	escalations
0	1	1	0	0
1	2	1	2	1
2	0	0	0	0
3	2	1	0	0
4	1	0	2	1

```
In [254... sns.heatmap(df_encoded.corr(), annot=True)
plt.show()
```



```
In [262... #heatmap using matplotlib- it is little bit difficult to plot that's why we
```

```
df_encoded = df_visual[
    ['plan_type', 'contract_type', 'churn_score',
     'churn_flag', 'churn_risk', 'escalations']
].copy()

# Ordinal encoding
order_mappings = {
    'plan_type': ['Basic', 'Standard', 'Premium'],
    'contract_type': ['Monthly', 'Annual'],
    'churn_risk': ['low', 'medium', 'high']
}

for col, order in order_mappings.items():
    df_encoded[col] = pd.Categorical(
```

```

        df_encoded[col],
        categories=order,
        ordered=True
    ).codes

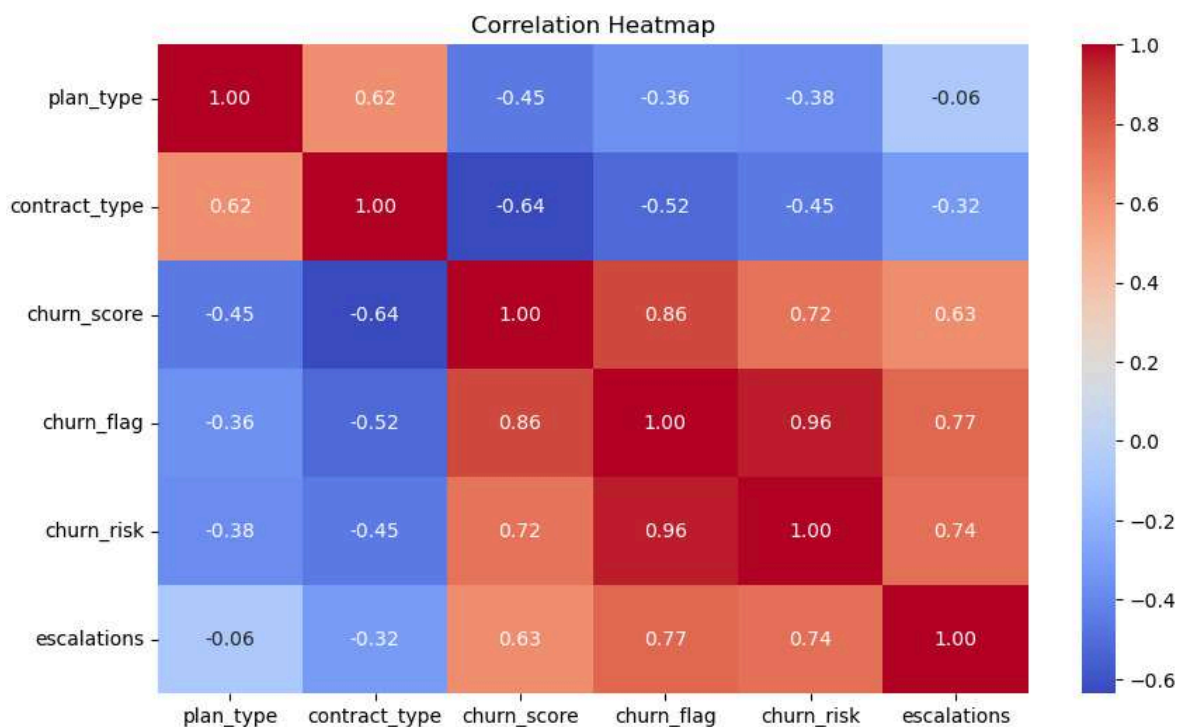
# Escalations: N = 0, Y = 1, NaN = 0
df_encoded['escalations'] = (
    df_encoded['escalations']
    .replace({'N': 0, 'Y': 1})
    .fillna(0)
    .astype(int)
)

# Heatmap
plt.figure(figsize=(10, 6))

sns.heatmap(
    df_encoded.corr(numeric_only=True),
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title('Correlation Heatmap')
plt.show()

```



```

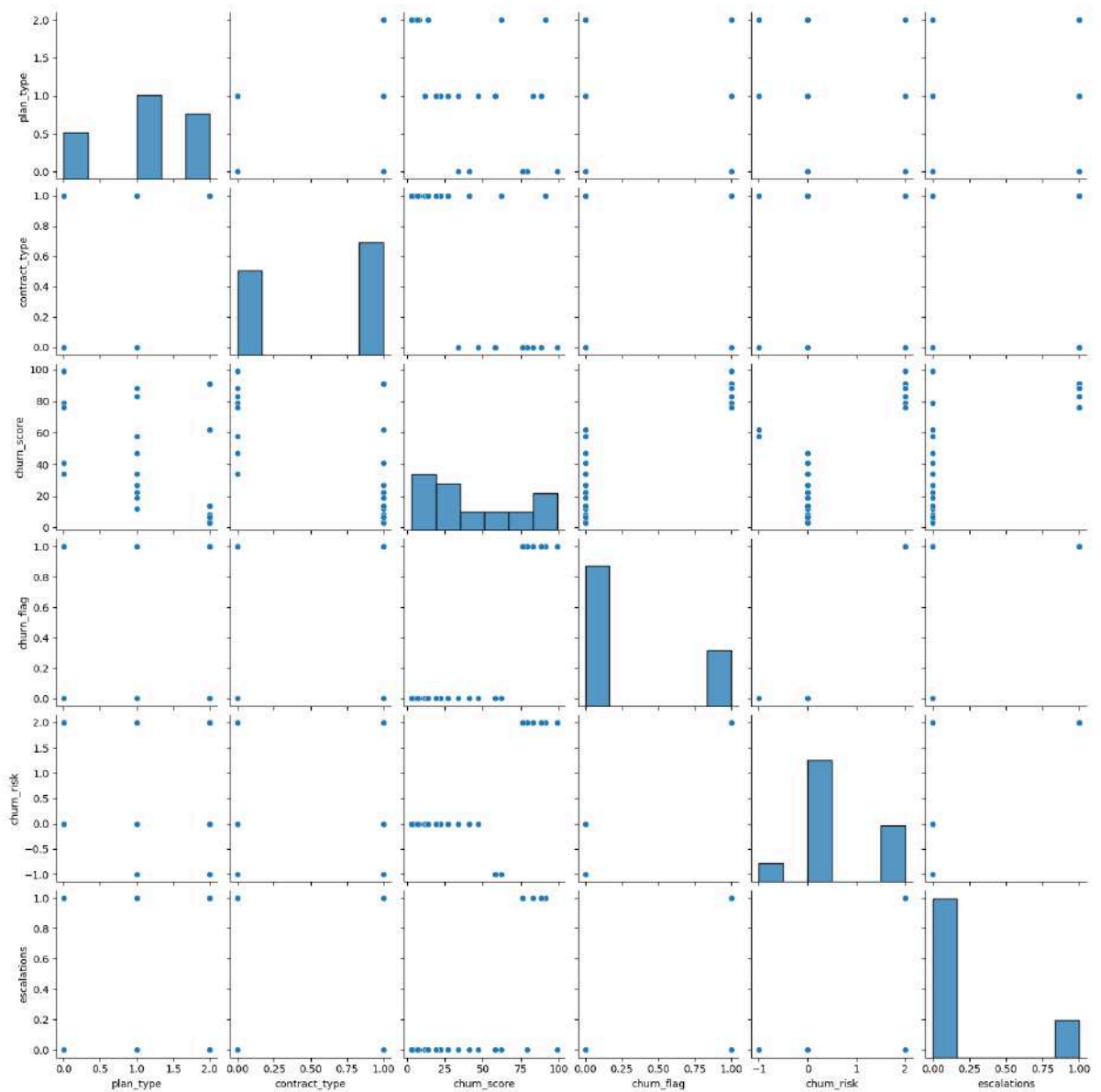
In [265... # pairplot -relationship in dataset
sns.pairplot(df_encoded)

```

```

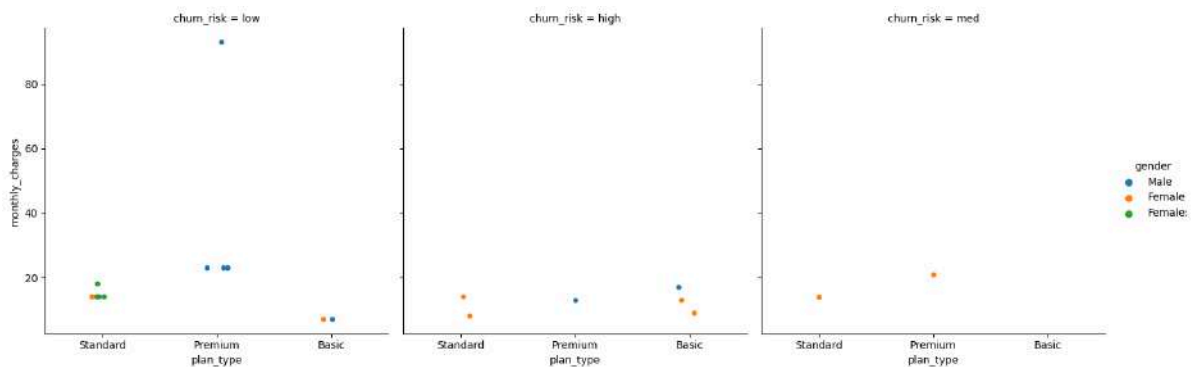
Out[265]: <seaborn.axisgrid.PairGrid at 0x134f932d0>

```



```
In [269...] sns.catplot(
    data=df_visual,
    x='plan_type',
    y='monthly_charges',
    hue='gender',
    col='churn_risk'
)

plt.show()
```



```
In [271...] pd.pivot_table(
    df_visual,
    index='plan_type',
    values='churn_flag',
```

```
aggfunc='mean'
).reset_index()
```

Out[271]:

	plan_type	churn_flag
0	Basic	0.600000
1	Premium	0.142857
2	Standard	0.222222

In [272... df_visual.columns

Out[272]:

```
Index(['customerid', 'subscription_start_date', 'subscription_type',
      'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
      'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
      'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
      'complaint_date', 'escalations', 'csat_score', 'complaint_date ',
      'complaint_count', 'tenure_days', 'escalation_flag', 'churn_risk',
      'cancellation_month'],
      dtype='object')
```

In [274... pd.pivot_table(

```
df_visual,
index='plan_type',
values=['monthly_charges', 'customerid', 'churn_flag'],
aggfunc={
    'monthly_charges': 'sum',
    'customerid': 'nunique',
    'churn_flag': 'mean'
})
).reset_index()
```

Out[274]:

	plan_type	churn_flag	customerid	monthly_charges
0	Basic	0.600000	5	52.95
1	Premium	0.142857	7	218.93
2	Standard	0.222222	9	123.91

In [276... *# working with sql in python (pandas)*

```
#create a table in sql
conn = sqlite3.connect('test_database.sqlite')

# create table
conn.execute("create table users(first_name text, country text,budget integer)")

#commit and save

conn.commit()
```

In [279... *# Insert data*

```
cursor = conn.cursor()

cursor.execute("""
INSERT INTO users (first_name, country, budget)
VALUES
    ('madhav', 'india', 5000),
    ('roshan', 'nepal', 7000),
    ('ruchi', 'india', 5000)
""")
```

```
# Commit and save
conn.commit()

print("Data inserted successfully")
```

Data inserted successfully

```
In [281]: # check inserted data into table
conn = sqlite3.connect('test_database.sqlite')
query = """select * from users"""

df_results = pd.read_sql(query,conn)
df_results.head()
```

```
Out[281]:
```

	first_name	country	budget
0	madhav	india	5000
1	roshan	nepal	7000
2	ruchi	india	5000

```
In [282]: # aggregation

query = """select country , sum(budget) as total_budget
from users
group by country

"""

df_agg = pd.read_sql(query,conn)
df_agg
```

```
Out[282]:
```

	country	total_budget
0	india	10000
1	nepal	7000

```
In [283]: # always close the conn with db once the task is over
conn.close()
```

```
In [ ]:
```